

Sample Paper 4 — Answer Key

Total: 60 marks (45×1 mark + 5×3 marks), 60 minutes.

Section 1 — Logical Reasoning

1. **D) Carrot** — Rose, Lily, and Lotus are flowers; Carrot is a vegetable.
2. **B) Account** — A key opens a lock; a password opens (access to) an account.
3. **A) HFU** — The rule is "+5 letters": $C \rightarrow H$, $A \rightarrow F$, $P \rightarrow U$.
4. **C) 144** — Each term doubles: 9, 18, 36, 72, 144.
5. **B) 13th** — Position from the right = $28 - 16 + 1 = 13$.
6. **C) East** — Two 90° clockwise turns from West = 180° turn = East.
7. **A) 1, 2, 4, 3** — A butterfly's life cycle: Egg $\rightarrow$ Caterpillar $\rightarrow$ Cocoon $\rightarrow$ Butterfly.
8. **C) 25** — 14, 21, 35 are multiples of 7; 25 is not.
9. **B) Friday** — 40 days = 5 full weeks (35 days, same day: Sunday) + 5 more days $\rightarrow$ Mon, Tue, Wed, Thu, Fri.
10. **B) Uncle** — Priya's mother's sister is her maternal aunt; the aunt's husband is Priya's uncle.
11. **C) 95** — The pattern is "double and add 1": $5 \times 2 + 1 = 11$, $11 \times 2 + 1 = 23$, $23 \times 2 + 1 = 47$, $47 \times 2 + 1 = 95$.
12. **A) RNCPV** — The rule is "+2 letters": $P \rightarrow R$, $L \rightarrow N$, $A \rightarrow C$, $N \rightarrow P$, $T \rightarrow V$.
13. **C) Sphere** — Circle, Oval, and Square are flat (2-D) shapes; a Sphere is a solid (3-D) shape.
14. **D) W** — $W > X > Y > Z$, so W is the heaviest.
15. **D) 21** — Each term is the sum of the two before it (a Fibonacci-style pattern): $8 + 13 = 21$.

Section 2 — Mathematical Reasoning

16. **A) 95,245** — $78,456 + 16,789 = 95,245$.
17. **A) 32,346** — $60,004 - 27,658 = 32,346$.
18. **A) 30,312** — $3,789 \times 8 = 30,312$.
19. **A) 8,554** — $68,432 \div 8 = 8,554$.
20. **C) 20** — HCF of 40 ($2^3 \times 5$) and 60 ($2^2 \times 3 \times 5$) is $2^2 \times 5 = 20$.
21. **A) 30** — Multiples of 10: 10, 20, 30... Multiples of 15: 15, 30... First common one = 30.
22. **A) 14/20** — Multiply top and bottom of $7/10$ by 2.
23. **A) 7/9** — Same denominator: $4 + 2 + 1 = 7$, so $7/9$.
24. **A) 8/25** — $0.32 = 32/100$, which simplifies ($\div 4$) to $8/25$.
25. **C) 60** — 40% of 150 = $0.40 \times 150 = 60$.

26. **B) 55%** — $11/20 = 0.55 = 55\%$.
27. **C) 45** — 1 part = $35 \div 7 = 5$; larger = 9 parts = 45.
28. **B) 25%** — Profit = $750 - 600 = ₹150$; profit % = $150/600 \times 100 = 25\%$.
29. **C) ₹1,680** — SI = $(7000 \times 4 \times 6) \div 100 = ₹1,680$.
30. **A) 6,045** — 6 kg = 6,000 g; + 45 g = 6,045 g.
31. **C) 3 h 45 min** — $225 \div 60 = 3$ remainder 45.
32. **B) 96 cm** — Perimeter = $2 \times (31 + 17) = 96$ cm.
33. **B) 169 sq cm** — Area = $13 \times 13 = 169$ sq cm.
34. **D) Countless (infinite)** — A circle can be folded along any diameter, so it has infinitely many lines of symmetry.
35. **C) 8** — Two full turns = 720° ; $720 \div 90 = 8$.

Section 3 — Everyday Mathematics

36. **C) 1,10,000** — $2,750 \times 40 = 1,10,000$.
37. **B) ₹2,860** — $7200 - 2450 - 1890 = 2,860$.
38. **C) 360 L** — $5/8$ of 960 = 600 L used; $960 - 600 = 360$ L left.
39. **C) ₹50** — $1,600 \div 32 = 50$.
40. **B) 322** — $14 \times 23 = 322$.
41. **C) ₹1,800** — SI = $(4500 \times 8 \times 5) \div 100 = ₹1,800$.
42. **B) 15%** — Profit = $2415 - 2100 = ₹315$; profit % = $315/2100 \times 100 = 15\%$.
43. **C) 51 km** — 2 h 50 min = $17/6$ hours; $18 \times 17/6 = 51$ km.
44. **B) 7 cups** — $7/8 \times 8 = 7$ cups.
45. **C) 168** — 42% of 400 = $0.42 \times 400 = 168$.

Section 4 — Achievers Section (3 marks each)

46. **C) 600** — If the original number is x : $0.65x = 390 \rightarrow x = 600$.
47. **C) 28** — Four consecutive numbers $n, n+1, n+2, n+3$ sum to $4n+6 = 106 \rightarrow n = 25$; the largest is $25+3 = 28$.
48. **C) ₹9,000** — SI = $(P \times 5 \times 6) \div 100 = 0.3 \times P = 2,700 \rightarrow P = 2,700 \div 0.3 = ₹9,000$.
49. **B) ₹2,000** — After clothes, $5/8$ of the money remains. After books ($1/5$ of that spent), $4/5$ of $5/8 = 1/2$ of the original remains. So $1/2$ of the total = ₹1,000 $\rightarrow$ total = ₹2,000.
50. **B) 40** — Let the numbers be $5x$ and $7x$. $(5x+8)/(7x+8) = 3/4 \rightarrow 4(5x+8) = 3(7x+8) \rightarrow 20x+32 = 21x+24 \rightarrow x = 8$. Smaller number = $5 \times 8 = 40$.
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Scoring guide (out of 60): 42+ is Level-2 cutoff range — strong. 30–41 is solid Level-1 performance. Below 30 — revisit the relevant topics before attempting Sample Paper 5.